

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **8**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 8

Consider the following polynomial function.

$$f(x) = -(x - 1)(x + 2)(x + 1)^2$$

Answer the questions regarding the graph of f .
Then, use this information to graph the function.

(a) Choose the end behavior of the graph of f .

(Blank 1,)

Blank 1, Options

- Falls to the left and rises to the right
- Rises to the left and falls to the right
- Rises to the left and rises to the right
- Falls to the left and falls to the right

(b) List each real zero of f according to the behavior of the graph at the x -axis near that zero. If there is more than one answer, separate them with commas. If there is no answer, write "None".

Zero(s) where the graph crosses the x -axis:

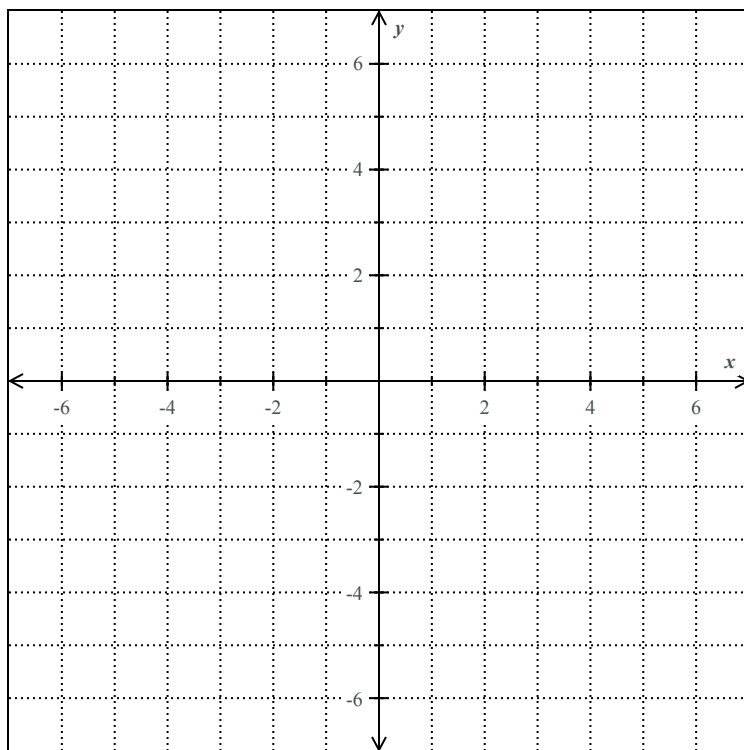
Zero(s) where the graph touches, but does not cross the x -axis:

(c) Find the y -intercept of the graph of f .

y -intercept:

(d) Graph $f(x) = -(x - 1)(x + 2)(x + 1)^2$ by doing the following.

- Plot all points where the graph of f intersects the x -axis or y -axis.
- For each point on the x -axis, select the correct behavior.
- Connect the points following the behaviors.



Question 2 of 8

For the polynomial below, -2 is a zero.

$$h(x) = x^3 - 10x - 12$$

Express $h(x)$ as a product of linear factors.

Question 3 of 8

The function below has at least one rational zero.
Use this fact to find *all* zeros of the function.

$$g(x) = 6x^3 - 31x^2 + 4x + 5$$

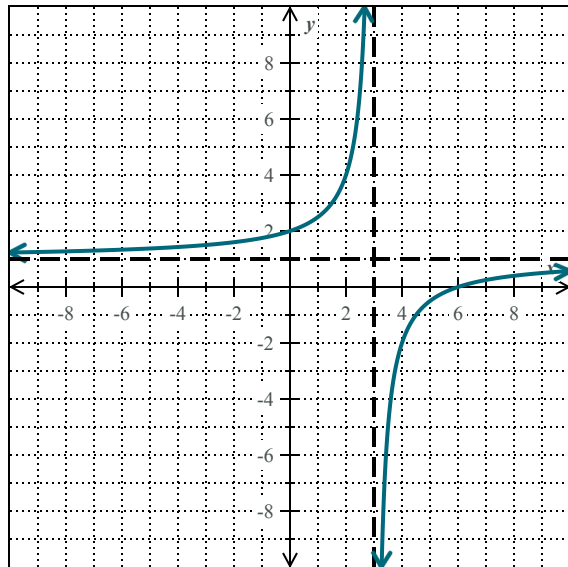
If there is more than one zero, separate them with commas. Write exact values, not decimal approximations.

Question 4 of 8

The graph of a rational function f is shown below.

Assume that all asymptotes and intercepts are shown and that the graph has no "holes".

Use the graph to complete the following.



(a) Write the equations for all vertical and horizontal asymptotes.

Vertical asymptote(s):

Horizontal asymptote(s):

(b) Find the domain and range of f .

Write each answer as an interval or union of intervals.

Domain:

Range:

(c) Find all x -intercepts and y -intercepts. Check *all* that apply.

x -intercept(s): ☐ 6 ☐ 2 ☐ 3 ☐ None

y -intercept(s): ☐ 1 ☐ 2 ☐ 6 ☐ None

(a) Write the equations for all vertical and horizontal asymptotes.

Vertical asymptote(s): _____

Horizontal asymptote(s): _____

(b) Find the domain and range of f .

Write each answer as an interval or union of intervals.

Domain: _____

Range: _____

(c) Find all x -intercepts and y -intercepts. Check *all* that apply.

x -intercept(s): ☐ 6 ☐ 2 ☐ 3 ☐ None

y -intercept(s): ☐ 1 ☐ 2 ☐ 6 ☐ None

Question 5 of 8

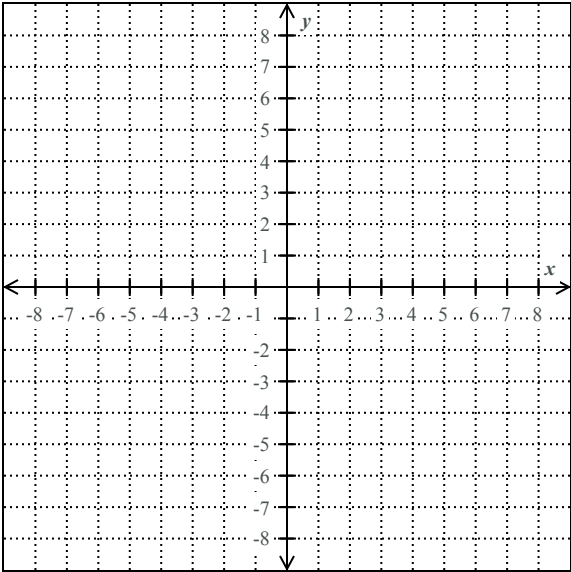
Graph all vertical and horizontal asymptotes of the rational function.

$$f(x) = \frac{2x^2 + 6}{2x - 3}$$

Question 6 of 8

Graph the rational function.

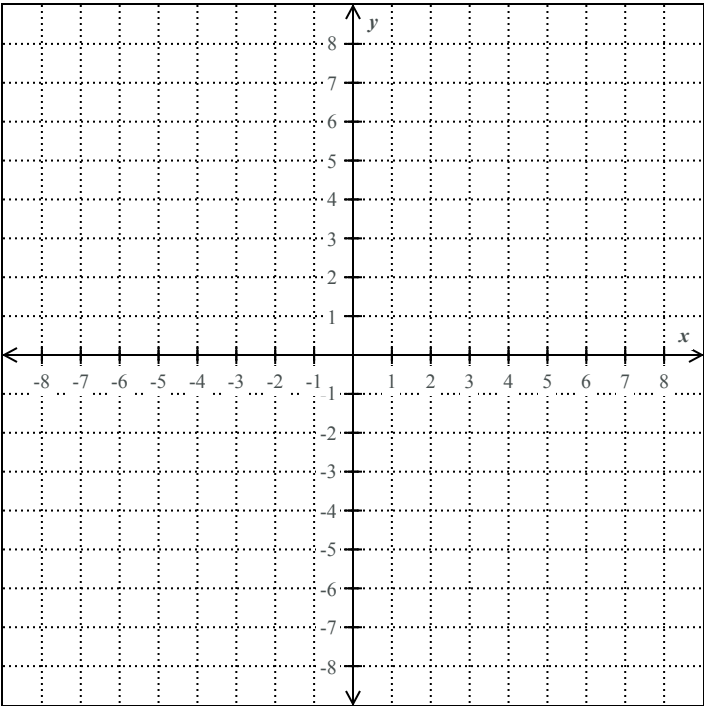
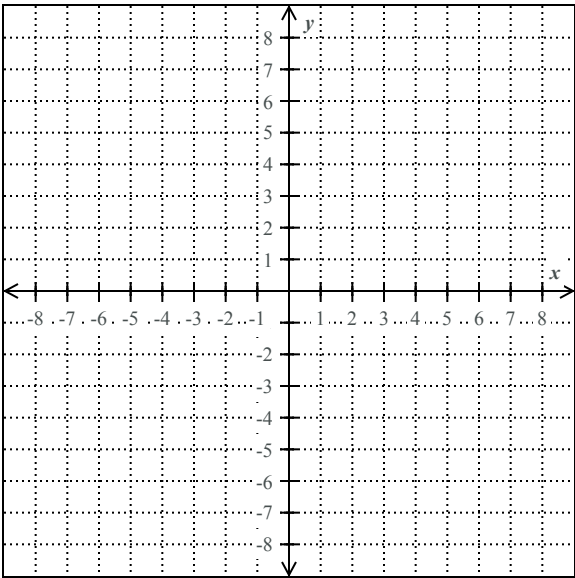
$$f(x) = \frac{-8}{x - 4}$$



Question 7 of 8

Graph the rational function.

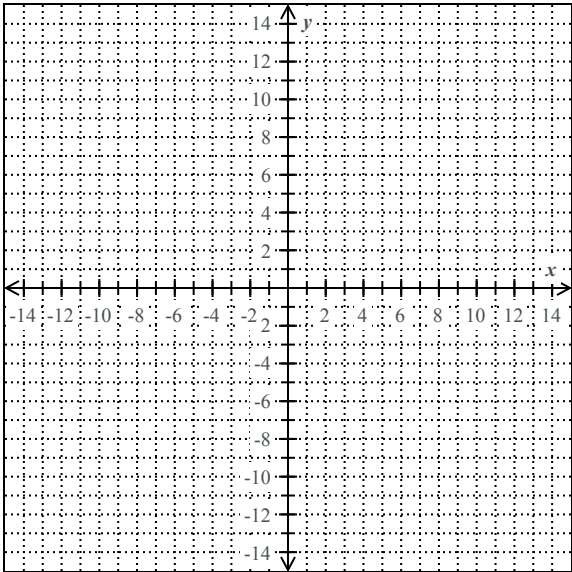
$$f(x) = \frac{2x - 2}{-x + 4}$$



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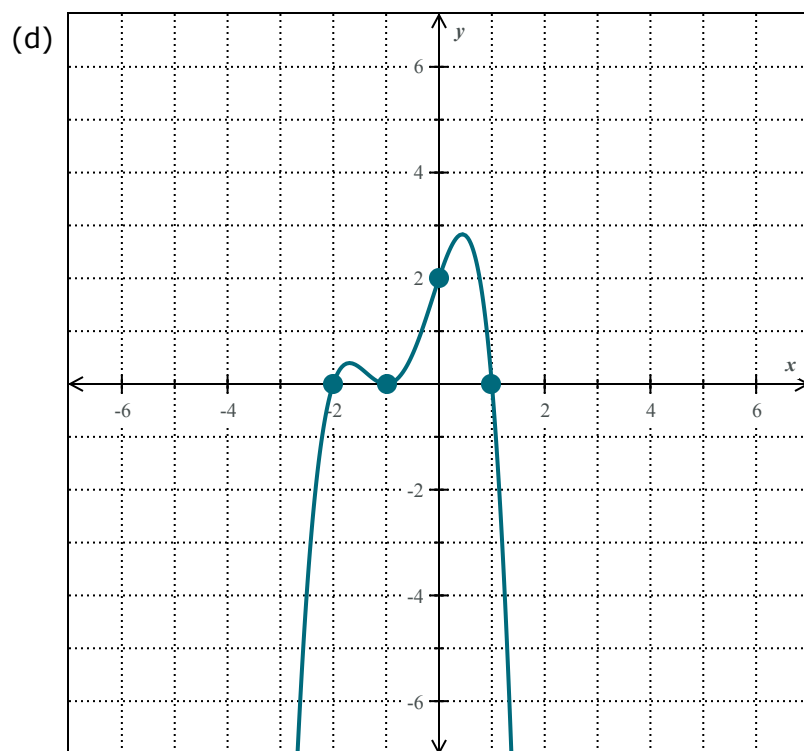
Graph the rational function.

$$f(x)=\frac{-10}{x^2-7x+10}$$



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(a) Falls to the left and falls to the right ▼

(b) Zero(s) where the graph crosses the x -axis: **1, -2**Zero(s) where the graph touches, but does not cross the x -axis: **-1**(c) y -intercept: **2****Question 2 of 8**

$$h(x) = (x + 2)(x - (1 + \sqrt{7}))(x - (1 - \sqrt{7}))$$

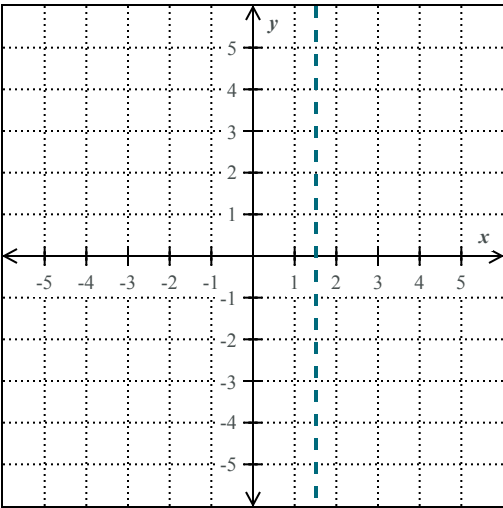
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$$5, \frac{1}{2}, -\frac{1}{3}$$

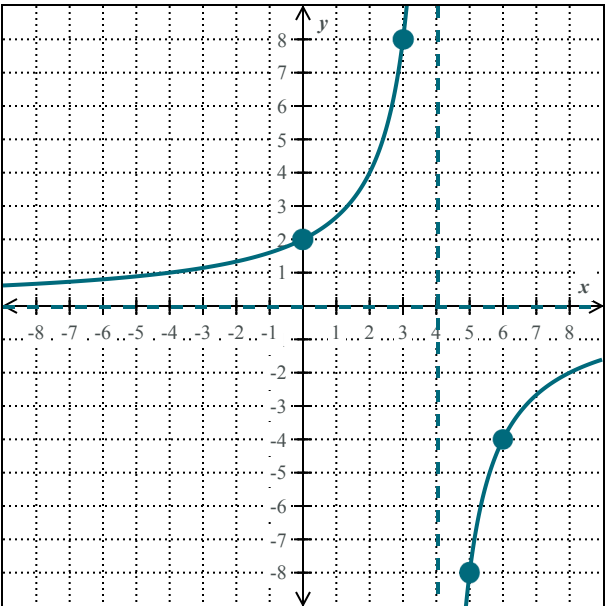
Question 4 of 8

- (a) Vertical asymptote(s): $x = 3$
Horizontal asymptote(s): $y = 1$
- (b) Domain: $(-\infty, 3) \cup (3, \infty)$
Range: $(-\infty, 1) \cup (1, \infty)$
- (c) x -intercept(s): ☒ 6 ☐ 2 ☐ 3 ☐ None
 y -intercept(s): ☐ 1 ☒ 2 ☐ 6 ☐ None

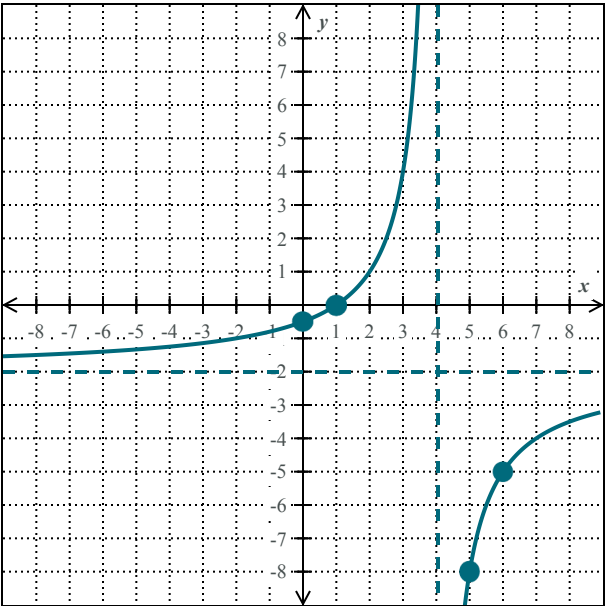
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Question 8 of 8

